1 ToDo - Next

- Mark: Github checken bzgl. Organisationseinheiten (OU) Account. Ziel Mitarbeiter haben Personalisierte Github accounts und können von unserer IT so lange sie contracted sind in unsere IBO Github Organisation eingehängt werden. Owner des REPO muss dan OU sein ??? To be discussed!
- all2Jonas: Bereitstellung von functions welche in einem Package konsolidiert werden sollen und alle anderen Mitarbeitern über OU zur Verfügung gestellt werden soll.
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2 Timeline

Laut Sitzung letzten Freitag wird das IBO noch für 2026 Matlab bezahlen, danach soll Weihnachten 2026 die Transition komplettiert sein. Ab Januar 2027 dann nahezu ausschliesslich Python/R, also Open Access Software im 'programming context'.
Heisst für unsere Arbeitsgruppe: Timeline abstecken.

- 2025-11: Auf mehreren Rechnern ist Python als Standard ausgerollt via Mark. Systeme und Python Package dependance file (oder wie das Ding nochmal heisst) wird auf feasability getested.
- 2026-01 Primäre Auswertung läuft über Python. Es sollen keine neuen Projekte mit Matlab begonnen werden.
- 2026-01 Info an Dozierende des TEC dass ab Winter 2026-27 Python gelehrt werden muss.
- 2027-01: Python läuft auf allen Maschinen mit Standardinstallation von Mark und 'alle' Projekte werden mit Python ausgewertet. Die Arbeitsgruppe stellt eine Grundauswertung zur Verfügung welche auf die unten stehende Architektur basiert (z.B. walking, running, vertical jumps)

3 Datei Architektur

`/json-struct`

- `MetaData` .... meta data
  - `PathFile: str` .... file path + name
  - `OriginalFiles: list[str]` .... list of all data source files
  - `Project: str` .... project name
  - `ProjectPI: str` .... project PI name
  - `SubjectID: str` .... subject ID
  - `Condition: str` .... condition name
  - `BodyMass: float` .... subject body mass [kg]
  - `BodyHeigt: float` .... subject body height [m]
  - `Sex: str` .... subject sex ('m' or 'f')
  - `Age: int`
  - `FileCreationLocal: str` .... datetime of file creation local time ('YYYY-MM-DD, HH:MM:SS.sss')
  - `FileCreationUTC: str` .... datetime of file creation UTC ('YYYY-MM-DD, HH:MM:SS.sss')
  - `LastUpdate: str` .... datetime of latest file update ('YYYY-MM-DD, HH:MM:SS.sss')
  - `Location: str` .... location of data collection
    - `Lat: float`
    - `Lon: float`
- `Trajectories` .... marker data
  - `SamplingFrequency: float` .... frame rate [Hz]
  - `NumFrames: int` .... total number of frames
  - `StartFrame: int` .... start frame of data if data was cropped
  - `EndFrame: int` .... end frame of data if data was cropped
  - `GlobalCoordinateSystem: str` .... description of each axis orientation
  - `Labeled` .... all labeled trajectories
    - `NumLabeled: int` .... number of labeled trajectories
    - `Data: ndarray(NumLabeled, 4, NumFrames)` .... data of labeled trajectories [x,y,z + residuals]

- Labels: `list[str]` ... labels of trajectories [length = NumLabeled]
- Type: `ndarray(NumLabeled, NumFrames)` ... virtual or measured [1 = measured; 0 = virtual]
- Unlabeled ... all unlabeled trajectories
  - NumUnlabeled: `int` ... number of labeled trajectories
  - Data: `ndarray(NumUnLabeled, 4, NumFrames)` ... data of labeled trajectories [x,y,z + residuals]
  - Type: `ndarray(NumUnLabeled, NumFrames)` ... virtual or measured [1 = measured; 0 = virtual]
- Analog
  - BoardName: `str` ... A/D board name
  - SamplingFrequency: `float` ... analog sampling frequency [Hz]
  - SamplingFactor: `float` ... factor SamplingFrequency/FrameRate
  - NumSamples: `int` ... total number of analog samples
  - Data: `ndarray(NumChannels, NumSamples)` ... analog data
  - Labels: `list[str]` ... analog channel names (length = NumChannels)
- ForcePlates: `list`
  - [0]
    - Name: `str` ... forceplate name
    - NumSamples: `int` ... total number of forceplate samples
    - SamplingFrequency: `float` ... forceplate sampling frequency [Hz]
    - SamplingFactor: `float` ... factor SamplingFrequency/FrameRate
    - Filter: `str` ... detailed description of filter, if applied. else: none
    - COP: `ndarray(3, NumSamples)` ... COP data
    - Force: `ndarray(3, NumSamples)` ... force data
    - Moment: `ndarray(3, NumSamples)` ... moment data
    - Location: `ndarray(3, 4, NumFrames)` ... forceplate corners over time
    - Position: `ndarray(4, NumFrames` ... forceplate origin + residuals
    - Rotation: `ndarray(3, 3, NumFrames)` ... forceplate rotation matrix
    - Offset: `ndarray(3,)` ... forceplate origin offset under corners
    - CoordinateSystem: `int` ... is forceplate data saved in (1 = global; 0 = local) coordinates
  - [1]
    - (same fields as [0])
- RigidBodies
  - [0]
    - Name: `str` ... rigid body name
    - Parent: `str` ... parent to which rigid body data is expressed to ('Global' or name)
    - TrajectoryNames: `list[str]` ... list of trajectory labels that define rigid body
    - NumFrames: `int` ... total number of frames
    - Filter: `str` ... detailed description of filter, if applied. else: none
    - Position: `ndarray(4, NumFrames)` ... force sensor origin + residuals
    - Rotation: `ndarray(3, 3, NumFrames)` ... force sensor rotation matrix
    - RPYs: `ndarray(3, NumFrames)` ... Roll, Ptich, Yaw
  - [1]
    - (same fields as [0])
- Events: `list`
  - [0]
    - Name: `str`
    - Frame: `int`
    - Time: `float`
  - [1]
    - (same fields as [0])
- CustomFields

4 Mindmap

- Converter (c3d -¿ .ibo-json; ...)
- Importer (.ibo-json -¿ python)

- crop/trim
- filter
- normalization
- visualization
- quality check
- subject descriptive statistics